Example 2.16 An alloy contains 26 % of copper. What quantity of alloy is required to get 260 g of copper ?

Solution:

Let the quantity of alloy required be Q g

Then 26 % of Q =260 g

$$\frac{26}{100} \times Q = 260\ g$$

$$Q = \frac{260 \times 100}{26} g$$

$$Q = \frac{26000}{26} g$$

$$Q = 1000\ g$$

Therefore, the required quantity of alloy is 1000 g.

Ratios as percentage

Sometimes ingredients used to prepare food can be represented in the form of ratio. Let us see some examples.

Example 2.17 Kuzhal's mother makes dosa by mixing the batter made from 1 portion of Urad dhal with 4 portions of rice. Represent each of the ingredients used in the batter as percentage.

Solution

Representing ingredients used in the batter as ratio, we get, rice : urad dhall $= 4:1$

Now, the total number of parts is $4+1=5$.

That is, $\frac{4}{5}$ portion of rice is mixed with $\frac{1}{5}$ portion of urad dhall.

Thus, the percentage of rice would be $\frac{4}{5} \times 100\% = \frac{400}{5}\% = 80\%$

The percentage of urad dhall would be $\frac{1}{5} \times 100\% = \frac{100}{5}\% = 20\%$

Example 2.18 A family cleans a house for pongal celebration by dividing the work in the ratio 1:2:3. Express each portion of work as percentage.

Solution

The total number of parts of the work is $1+2+3=6$

That is, the work is divided into 3 portions as $\frac{1}{6}, \frac{2}{6}$ and $\frac{3}{6}$.

Thus, the percentage of $\frac{1}{6}^{th}$ portion of work would be $\frac{1}{6} \times 100\% = \frac{100}{6}\% = 16\frac{2}{3}\%$

Similarly, the percentage of $\frac{2}{6}^{th}$ portion of work would be $\frac{2}{6} \times 100\% = \frac{200}{6}\% = 33\frac{1}{3}\%$

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